

Program : Diploma in Instrumentation Engineering	
Course Code : 6081A	Course Title: Advanced Biomedical Instrumentation
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week : 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To appreciate the need for Biomedical Instrumentation
- To learn medical assistance techniques and Therapeutic Instruments
- To outline the Imaging Techniques
- To impart the knowledge of Biotelemetry and need for Safety of Medical Instruments

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics of sensors and transducers	3082	Sensors & Transducers	3
Basics of biomedical instruments	4084	Analytical and Biomedical Instruments	4

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain Biomedical Instrumentation system and Bio-potential Electrodes	10	Understanding
CO2	Explain various therapeutic equipments.	16	Understanding
CO3	Describe the principle and working of imaging techniques such as X-ray, CT, MRI, Thermal and Ultrasound	16	Understanding
CO4	Describe patient monitoring, biotelemetry and safety systems related to biomedical instrumentation.	16	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						3
CO4	2				2		2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain Biomedical Instrumentation system and Bio-potential Electrodes		
M1.01	Describe the Block diagram of Biomedical Instrumentation system.	1	Understanding
M1.02	Describe the analytical, diagnostic, therapeutic and surgical types of Biomedical Instruments.	1	Remembering
M1.03	List the Electrical classification of Biomedical Instruments.		Remembering
M1.04	Describe the role of Biomedical equipment technician.	1	Remembering
M1.05	Define biometrics.		Remembering
M1.06	Explain the sources of Biomedical signals	1	Understanding
M1.07	Explain the problems encountered while performing Measurement of living organisms.	1	Understanding
M1.08	List different types of bioelectric potentials.	1	Remembering
M1.09	Define ECG, EEG, EMG, ERG, EOG and EGG		Remembering
M1.10	Define electrode.	2	Remembering
M1.11	Name the three types of Bio-potential electrodes.		Remembering
M1.12	Explain microelectrodes, skin surface electrodes and needle electrodes.		Understanding
M1.13	Recall the principle and working of different transducers used for biomedical applications.	2	Remembering
M1.14	Explain the criteria for selecting biomedical transducers.		Understanding

Contents:

Overview of Biomedical Instrumentation system: Block diagram - Types of biomedical instruments: Analytical, Diagnostic, Therapeutic and Surgical Instruments - Electrical classification of Biomedical Instruments.

Role of Biomedical equipment technician - Development of Biomedical Instrumentation, biometrics - Sources of Biomedical signals - Problems encountered while performing Measurement of living organisms.

Bioelectric potentials - examples (ECG, EEG, EMG, ERG, EOG, EGG, etc.) - Bio potential electrodes – theory – microelectrodes - skin surface electrodes - needle electrodes

Transducers for biomedical applications (brief discussion of basic principles of force, displacement, velocity, acceleration and pressure transducers). Criteria for selecting biomedical transducers

CO2	Explain various therapeutic equipments.		
M2.01	State the need of pacemakers	3	Understanding
M2.02	Classify different types of pacemakers		Remembering
M2.03	Compare the implantable pacemakers and external pacemakers		Remembering
M2.04	Draw the block diagram of a ventricular synchronous demand pacemaker and explain its operation		Remembering
M2.05	State the need of defibrillators.	2	Remembering
M2.06	Describe ac defibrillators and dc defibrillators.		Understanding
M2.07	Explain the functions of haemodialysis machine.	2	Understanding
M2.08	Describe the working of haemodialysis machine.		Understanding
M2.09	State the use of respirators.	1	Remembering
M2.10	Explain the working of respirators		Understanding
M2.11	State the need of ventilator.	3	Remembering
M2.12	List the types of ventilators.		Remembering
M2.13	Explain the classification of ventilators.		Understanding
M2.14	Draw the block diagram of a microprocessor		Understanding

	controlled ventilator and explain its operation		
M2.15	List the different types of diathermy equipments.	3	Remembering
M2.16	Explain the working of short wave diathermy unit		Remembering
M2.17	List the advantages and disadvantages of short-wave diathermy treatment.		Understanding
M2.18	Explain the working of Ultrasonic diathermy unit		Remembering
M2.19	Explain the necessity of blood cell counting	2	Understanding
M2.20	Explain the working of electrical conductivity blood cell counter		Understanding
	Series Test – I	1	

Contents:

Pacemakers - types of pacemakers - Implantable pacemakers and external pacemakers - ventricular synchronous demand pacemaker – defibrillators - ac defibrillators and dc defibrillators - haemodialysis machine – Respirators – ventilator - diathermy equipment - short wave diathermy unit - Advantages and disadvantages of short-wave diathermy treatment - Ultrasonic diathermy unit - electrical conductivity blood cell counter

CO3	Describe the principle and working of imaging techniques such as X-ray, CT, MRI, Thermal and Ultrasound		
M3.01	List the properties of X-rays.	1	Remembering
M3.02	Explain the principle of generation of X-ray		Understanding
M3.03	Explain the construction and operation of an X-ray machine with block diagram.	3	Understanding
M3.04	Describe the visualization of X-ray.		Understanding
M3.05	Explain Dental X-ray machines	1	Understanding
M3.06	Describe portable and mobile X-ray units		Understanding
M3.07	List two limitations of conventional X-rays	3	Remembering
M3.08	Describe the basic principle of Computed Tomography.		Understanding
M3.09	Explain the operation of CT scanner with block diagram.		Understanding
M3.10	Describe the working of Nuclear Magnetic Resonance Imaging system	3	Understanding

M3.11	Explain the biological effects of NMR imaging		Understanding
M3.12	List the advantages of NMR imaging system		Remembering
M3.13	Compare X-ray, Nuclear medical imaging and ultrasound		1
M3.14	List the properties of ultrasonic waves	Remembering	
M3.15	Explain applications of ultrasound in medical field	Understanding	
M3.16	Explain the working of a basic pulse-echo system	4	Understanding
M3.17	Explain A-scan, M-scan and B-scan in an Ultrasonic imaging system.		Understanding
Contents: X-rays: Properties of x-rays - Principle of generation, x-ray machine, visualization of x-rays – Dental x-ray machine - portable and mobile x-ray units Computed Tomography: Basic Principle - working of CT scanner Nuclear Magnetic Resonance Imaging system: Basic NMR components – Biological effects of NMR Imaging – Advantages of NMR Imaging system Ultrasonic Imaging Systems: Compare X-ray, Nuclear medical imaging and ultrasound – Properties of ultrasonic waves – Medical ultrasound – Pulse-echo apparatus – A-scan, M-scan and B-scan			
CO4	Describe patient monitoring, biotelemetry and safety systems related to biomedical instrumentation.		
M4.01	Sate the objectives of patient monitoring system.	6	Remembering
M4.02	Explain the working of cardioscope.		Understanding
M4.03	Describe the working of a bedside monitor.		Understanding
M4.04	Explain the functions of central station monitors.		Understanding
M4.05	Define biotelemetry	3	Remembering
M4.06	State the need of biotelemetry.		Remembering
M4.07	Explain the block diagram of biotelemetry system.		Understanding
M4.08	Describe the applications of biotelemetry		Understanding

M4.09	Define telemedicine.	3	Remembering
M4.10	List the applications of telemedicine.		Remembering
M4.11	Name the essential parameters for telemedicine.		Remembering
M4.12	Explain telemedicine using video conferencing.		Understanding
M4.13	Explain the principle of telemedicine using mobile communication with the help of a block diagram.		Understanding
M4.14	Explain the effects of electric current on the human body.	4	Understanding
M4.15	Define micro shock and macro shock.		Remembering
M4.16	Summarize the safety codes for electro-medical equipment.		Remembering
M4.17	List the precautions to minimize electric shock hazards.		Remembering
	Series Test – II	1	
Contents: Patient monitoring system: Cardiac monitor (Cardioscope), Bed-side monitors, Central station monitors Biotelemetry - Block diagram of biotelemetry system - Applications of biotelemetry – Telemedicine: video conferencing, Telemedicine using mobile communication, Applications of telemedicine Electrical safety – effects of electric current on the human body, micro and macro shock hazards - safety codes for electro-medical equipment – Precautions to minimize electric shock hazards			

Text / Reference:

T/R	Book Title / Author
T1	Hand Book of Biomedical Instrumentation, R. S. Kandpur - Tata McGraw-Hill.
T2	Biomedical Instrumentation and Measurements, Leslie Cromwell & Free. J.Weibell – Prentice Hall.
R1	Medical Instrumentation, John. G. Webster - Wiley India.
R2	Biomedical Instruments, Dr. Arumugham - Anuradha publications.

Online Resources:

Sl. No	Website Link
1	https://www.electrical4u.com/action-potential-and-resting-potential
2	https://en.wikipedia.org/wiki/Electrocardiography
3	https://en.wikipedia.org/wiki/Artificial_cardiac_pacemaker